

Linear Models Lecture 11: Asymptotic theory for OLS

Robert Gulotty

University of Chicago

April 24, 2026

Where We Left Off

Last lecture. Stated the CLT:

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

True for any iid sample with finite variance; we did not prove it.

Today. The proof requires one tool we have not developed: the **moment generating function**. We will:

- 1 Define the MGF and work a complete example.
- 2 Sketch the CLT proof in three steps.
- 3 Turn to the delta method and OLS asymptotics.

What Is a Moment Generating Function?

Definition (Moment generating function)

For a random variable X ,

$$M_X(t) \equiv \mathbb{E}[e^{tX}] = \int e^{tx} f(x) dx,$$

defined for t in some open interval around 0 where the expectation is finite.

Two facts make the MGF useful.

- 1 Its derivatives at 0 are the moments: $M_X^{(k)}(0) = \mathbb{E}[X^k]$ (next slide).
- 2 For independent X and Y , $M_{X+Y}(t) = M_X(t) M_Y(t)$.

Example: The Distribution of a Sum (Two Fair Dice)

Setup. Independent rolls X, Y , each uniform on $\{1, 2, \dots, 6\}$. What is the pmf of $X + Y$?

Count pairs. Each of the 36 pairs (x, y) is equally likely. For each possible sum s , count the pairs with $x + y = s$:

s	2	3	4	5	6	7	8	9	10	11	12
#pairs	1	2	3	4	5	6	5	4	3	2	1

So $p_{X+Y}(s) = \text{\#pairs}/36$. Triangular, peaked at 7.

The formula. The count at s is the number of ways $X = x$ can pair with $Y = s - x$, weighted by their joint pmf:

$$p_{X+Y}(s) = \sum_x p_X(x) p_Y(s - x).$$

This is the **convolution** of the two pmfs. For continuous X, Y , replace the sum with an integral.

Why the Product Property Matters

For continuous X, Y , the convolution is an integral with the same structure:

$$f_{X+Y}(s) = \int f_X(x) f_Y(s-x) dx.$$

This integral is awkward to compute once, and worse to iterate n times for a sum $X_1 + \cdots + X_n$.

In MGF terms, the same computation is just a product:

$$M_{X_1+\cdots+X_n}(t) = M_{X_1}(t) \cdots M_{X_n}(t).$$

The CLT proof exploits this at every step.

Why “Moment Generating”? (1 of 2)

Taylor-expand e^{tX} in powers of t :

$$e^{tX} = 1 + tX + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \dots$$

Take expectations term by term (valid whenever the MGF exists near 0):

$$M_X(t) = \mathbb{E}[e^{tX}] = 1 + t\mathbb{E}[X] + \frac{t^2}{2!}\mathbb{E}[X^2] + \frac{t^3}{3!}\mathbb{E}[X^3] + \dots$$

This is a power series in t whose coefficients are the moments $\mathbb{E}[X^k]/k!$.

Why “Moment Generating”? (2 of 2)

Compare to the ordinary Taylor series of M_X at 0:

$$M_X(t) = M_X(0) + M'_X(0)t + \frac{M''_X(0)}{2!}t^2 + \frac{M'''_X(0)}{3!}t^3 + \dots$$

Two power series in t for the same function must have equal coefficients. Match them:

$$M_X^{(k)}(0) = \mathbb{E}[X^k] \text{ for every } k \geq 0.$$

Differentiating M_X at 0 pulls out the moment of order k — that is why “moment generating.”
 One analytic function encodes the entire sequence of moments.

Worked Example: Exponential(λ)

Setup. $X \sim \text{Exp}(\lambda)$, density $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$. We know $\mathbb{E}[X] = 1/\lambda$. Recover it from the MGF.

Step 1 — Compute the MGF.

$$M_X(t) = \int_0^\infty e^{tx} \lambda e^{-\lambda x} dx = \lambda \int_0^\infty e^{-(\lambda-t)x} dx = \frac{\lambda}{\lambda - t}, \quad t < \lambda.$$

Step 2 — Differentiate at 0.

$$M'_X(t) = \frac{\lambda}{(\lambda - t)^2} \implies M'_X(0) = \frac{1}{\lambda} = \mathbb{E}[X]. \quad \checkmark$$

No integration of $x f(x)$: we computed the MGF once and read the mean off the derivative at 0. Higher moments work the same way.

Worked Example: Standard Normal

Setup. $Z \sim N(0, 1)$ has MGF (derive once and commit to memory)

$$M_Z(t) = \mathbb{E}[e^{tZ}] = e^{t^2/2}.$$

This is the target of the CLT proof. Every step of the proof tries to show $M_{Z_n}(t) \rightarrow e^{t^2/2}$.

Pull off moments by differentiation.

$$M'_Z(t) = t e^{t^2/2},$$

$$M'_Z(0) = 0 = \mathbb{E}[Z].$$

$$M''_Z(t) = (1 + t^2) e^{t^2/2},$$

$$M''_Z(0) = 1 = \mathbb{E}[Z^2].$$

$$M'''_Z(t) = (3t + t^3) e^{t^2/2},$$

$$M'''_Z(0) = 0 = \mathbb{E}[Z^3].$$

$$M^{(4)}_Z(t) = (3 + 6t^2 + t^4) e^{t^2/2},$$

$$M^{(4)}_Z(0) = 3 = \mathbb{E}[Z^4].$$

Odd moments vanish (symmetry); even moments are 1, 3, 15, 105, ... One function $e^{t^2/2}$ encodes all of them.

Why MGFs Are the Right Tool for Sums

The key property. If $X \perp Y$, then

$$M_{X+Y}(t) = \mathbb{E}[e^{t(X+Y)}] = \mathbb{E}[e^{tX} e^{tY}] \stackrel{\perp}{=} \mathbb{E}[e^{tX}] \mathbb{E}[e^{tY}] = M_X(t) M_Y(t).$$

For an iid sum $S_n = X_1 + \cdots + X_n$: $M_{S_n}(t) = [M_X(t)]^n$.

Compare the density domain. Adding independent RVs convolves densities, $f_{X+Y}(z) = \int f_X(x) f_Y(z-x) dx$ — a painful integral. In the MGF domain, a product.

This is why the CLT proof uses MGFs: sums $\rightarrow$ products $\rightarrow$ powers $\rightarrow$ something we can Taylor-expand.

The MGF Continuity Theorem

Theorem (Lévy continuity, MGF version)

Suppose $M_{X_n}(t) \rightarrow M_X(t)$ for every t in some open interval around 0, and M_X is the MGF of a proper distribution. Then $X_n \xrightarrow{d} X$.

Why we need this. Convergence of CDFs is hard to show directly; MGFs are analytic. The theorem lets us work at the level of MGFs and *conclude* convergence in distribution for free.

The plan for the CLT proof.

- 1 Standardize the sample mean as Z_n ; compute $M_{Z_n}(t)$ via the factorization property.
- 2 Show $M_{Z_n}(t) \rightarrow e^{t^2/2}$.
- 3 Continuity theorem $\Rightarrow Z_n \xrightarrow{d} N(0, 1)$.

Sketch of the CLT Proof: Setup

Goal. Show $M_{Z_n}(t) \rightarrow e^{t^2/2}$ for $Z_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$. The MGF continuity theorem then gives $Z_n \xrightarrow{d} N(0, 1)$.

Standardize. Let $W_i = (X_i - \mu)/\sigma$. Then

$$\mathbb{E}[W_i] = 0, \quad \mathbb{E}[W_i^2] = 1, \quad Z_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n W_i.$$

Factor via independence. Since the W_i are iid, the MGF of the sum factors (previous slide):

$$M_{Z_n}(t) = \mathbb{E} \left[\exp \left(\frac{t}{\sqrt{n}} \sum_i W_i \right) \right] = [M_W(t/\sqrt{n})]^n.$$

Sketch of the CLT Proof: Taylor-Expand the MGF

Taylor-expand $M_W(s)$ around $s = 0$:

$$M_W(s) = M_W(0) + M'_W(0)s + \frac{1}{2}M''_W(0)s^2 + O(s^3).$$

Evaluate the derivatives using the moment identities $M_W^{(k)}(0) = \mathbb{E}[W^k]$ (two slides back):

$$M_W(0) = 1, \quad M'_W(0) = \mathbb{E}[W] = 0, \quad M''_W(0) = \mathbb{E}[W^2] = 1.$$

Hence $M_W(s) = 1 + \frac{1}{2}s^2 + O(s^3)$.

Plug in $s = t/\sqrt{n}$:

$$M_W(t/\sqrt{n}) = 1 + \frac{t^2}{2n} + O(n^{-3/2}).$$

Sketch of the CLT Proof: Exponential Limit

Raise to the n -th power (independence):

$$M_{Z_n}(t) = [M_W(t/\sqrt{n})]^n = \left[1 + \frac{t^2/2}{n} + O(n^{-3/2})\right]^n.$$

Apply the exponential limit $(1 + x/n)^n \rightarrow e^x$. The $O(n^{-3/2})$ term vanishes faster than $1/n$, so it drops out:

$$M_{Z_n}(t) \longrightarrow e^{t^2/2}. \quad \blacksquare$$

The whole proof on one line. Taylor-expand the MGF, raise to the n -th power via independence, take the exponential limit, and invoke continuity.

Today's Claims

- The least squares estimator $\hat{\beta}$ is *consistent* for the projection coefficient β .
- There is an asymptotic normal approximation to the distribution of $\hat{\beta}$ (once normalized).
- Similarly, the error variance estimators $\hat{\sigma}^2$ and s^2 are consistent for σ^2 .
- There is a consistent estimator of $\mathbf{V}_\beta$, either under homoskedasticity $\hat{\mathbf{V}}_\beta^0$, or otherwise.
- We can use z-statistics for asymptotic standard errors, confidence intervals, and prediction intervals.

Strategy: Apply the tools from Lecture 10 — WLLN, CLT, CMT — plus the delta method (introduced today) to derive the asymptotic properties of OLS.

Motivation: Distribution of a Function of Estimators

- Suppose we want to estimate some function of parameters $f(b_1, b_2) = b_1/b_2$.
- What is the distribution of $\mathbf{f}(\mathbf{b})$?
- We can use the **Delta Method** (Asymptotic Normality + Taylor Expansion) to approximate it.
- The idea: if $\mathbf{b}$ is approximately normal (by the CLT), then a smooth function $\mathbf{f}(\mathbf{b})$ is also approximately normal, with variance determined by the derivatives of f .

Asymptotic Normality of $\mathbf{b}$ (Preview)

Claim (proved later this lecture): if errors are iid with mean zero and finite variance and $\mathbf{X}$ is full rank, the CLT gives:

$$\mathbf{b} \overset{a}{\sim} N \left[\boldsymbol{\beta}, \frac{\sigma^2}{n} \left[\text{plim} \frac{\mathbf{X}'\mathbf{X}}{n} \right]^{-1} \right], \quad \text{estimated by } s^2(\mathbf{X}'\mathbf{X})^{-1}$$

For now, take this as given. Once $\mathbf{b}$ is approximately normal, we can ask: what is the distribution of a *function* of $\mathbf{b}$?

The Delta Method: Linearization

The Jacobian. For a smooth $\mathbf{f} : \mathbb{R}^K \rightarrow \mathbb{R}^J$, define the $J \times K$ matrix of partial derivatives

$$\mathbf{C}(\mathbf{b}) = \frac{\partial \mathbf{f}(\mathbf{b})}{\partial \mathbf{b}'}$$

Evaluate at the truth. Consistency of $\mathbf{b}$ plus continuity of $\mathbf{C}$:

$$\text{plim } \mathbf{C}(\mathbf{b}) = \frac{\partial \mathbf{f}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}'} \equiv \boldsymbol{\Gamma}. \quad (\text{Slutsky})$$

First-order Taylor expansion from $\boldsymbol{\beta}$ to $\mathbf{b}$:

$$\mathbf{f}(\mathbf{b}) = \mathbf{f}(\boldsymbol{\beta}) + \boldsymbol{\Gamma}(\mathbf{b} - \boldsymbol{\beta}) + O_p(n^{-1}).$$

The Delta Method: Asymptotic Distribution

Combine the Taylor expansion with $\mathbf{b} \stackrel{a}{\sim} N(\boldsymbol{\beta}, \text{Asy. Var}(\mathbf{b}))$:

$$\mathbf{f}(\mathbf{b}) \stackrel{a}{\sim} N\left(\mathbf{f}(\boldsymbol{\beta}), \boldsymbol{\Gamma} [\text{Asy. Var}(\mathbf{b})] \boldsymbol{\Gamma}'\right).$$

Estimator. Plug in $\mathbf{C}(\hat{\boldsymbol{\beta}})$ and the estimated variance of $\hat{\boldsymbol{\beta}}$:

$$\widehat{\text{Asy. Var}}(\mathbf{f}(\hat{\boldsymbol{\beta}})) = \mathbf{C}(\hat{\boldsymbol{\beta}}) [s^2(\mathbf{X}'\mathbf{X})^{-1}] \mathbf{C}(\hat{\boldsymbol{\beta}})'$$

HW4 Q1(b) is the scalar case: $g(x) = e^x$ gives $[g'(\mu)]^2 \sigma^2 = \sigma^2 e^{2\mu}$.

Long-Run Effects with a Lagged DV (Greene 4.4)

Dynamic model: $y_t = \beta_0 + \beta_1 x_t + \gamma y_{t-1} + \epsilon_t$. Raise x_t by 1 and trace forward (holding the system at its long-run mean otherwise):

$$y_t^* = \bar{y} + \beta_1, \quad y_{t+1}^* = \bar{y} + \gamma\beta_1, \quad y_{t+2}^* = \bar{y} + \gamma^2\beta_1, \quad \dots$$

The cumulative long-run impact is a geometric series:

$$\beta_1 + \gamma\beta_1 + \gamma^2\beta_1 + \dots = \frac{\beta_1}{1 - \gamma}.$$

Equivalent shortcut: solve $\bar{y} = \beta_0 + \beta_1 \bar{x} + \gamma \bar{y}$ at long-run equilibrium.

Application: Long-Run Gasoline Elasticities

Log-log gasoline demand with lagged consumption,

$$\log(G/\text{pop})_t = \beta_1 + \beta_2 \log P_{G,t} + \beta_3 \log(Y/\text{pop})_t + \beta_4 \log(G/\text{pop})_{t-1} + e_t.$$

- Short-run price elasticity = β_2 ; short-run income elasticity = β_3 .
- By the lagged-DV iteration argument, long-run elasticities are

$$\frac{\beta_2}{1 - \beta_4} \quad (\text{price}), \quad \frac{\beta_3}{1 - \beta_4} \quad (\text{income}).$$

- Both are nonlinear in β — next slide gets their standard errors via the delta method.

Delta Method for Long-Run Elasticities: Partial

Stack the two long-run elasticities as a vector-valued function of $\beta = (\beta_1, \beta_2, \beta_3, \beta_4)'$:

$$f(\beta) = \begin{pmatrix} f_1(\beta) \\ f_2(\beta) \end{pmatrix} = \begin{pmatrix} \beta_2/(1 - \beta_4) \\ \beta_3/(1 - \beta_4) \end{pmatrix}.$$

Partial of $f_1 = \beta_2/(1 - \beta_4)$:

- $\partial f_1/\partial \beta_1 = 0$ — intercept does not enter.
- $\partial f_1/\partial \beta_2 = 1/(1 - \beta_4)$ — f_1 is linear in β_2 with that slope.
- $\partial f_1/\partial \beta_3 = 0$ — β_3 does not enter.
- $\partial f_1/\partial \beta_4 = \beta_2/(1 - \beta_4)^2$ — chain rule on $1/(1 - \beta_4)$.

Partial of $f_2 = \beta_3/(1 - \beta_4)$. Same pattern with the roles of β_2 and β_3 swapped: only $\partial f_2/\partial \beta_3$ and $\partial f_2/\partial \beta_4$ are nonzero.

Delta Method for Long-Run Elasticities: The Sandwich

Stack the partials into the 2×4 Jacobian (one row per function, one column per coefficient):

$$\mathbf{C}(\beta) = \frac{\partial \mathbf{f}}{\partial \beta'} = \begin{bmatrix} 0 & \frac{1}{1-\beta_4} & 0 & \frac{\beta_2}{(1-\beta_4)^2} \\ 0 & 0 & \frac{1}{1-\beta_4} & \frac{\beta_3}{(1-\beta_4)^2} \end{bmatrix}.$$

Plug in $\hat{\beta}$ and sandwich with the estimated variance of $\hat{\beta}$:

$$\widehat{\text{Var}}(\hat{\mathbf{f}}) = \mathbf{C}(\hat{\beta}) \hat{\mathbf{V}}_{\hat{\beta}} \mathbf{C}(\hat{\beta})'.$$

Result is 2×2 : diagonal entries are the variances of the long-run price and income elasticities, off-diagonal is their covariance.

Gas Consumption: Implementation

```
fm <- lm(logG ~ logGASP + log(INCOME) + log(PNC)
          + log(PUC) + logGlag, data = USGasG)

# By hand
bs <- coef(fm); V <- vcov(fm)
g <- c(0, 1/(1-bs[6]), 0, 0, 0, bs[2]/(1-bs[6])^2)
se <- sqrt(t(g) %*% V %*% g)

# Canned: car::deltaMethod(fm, "logGASP/(1-logGlag)")
# Estimate -0.411, SE 0.152, 95% CI [-0.710, -0.113]
# Stata: nlcom (_b[x2] / (1 - _b[x6]))
```

Long-run price elasticity $\approx -0.41 \pm 0.30$; income elasticity $\approx 0.97 \pm 0.32$ — both larger in magnitude than short-run.

Delta Method for Probit: Average Marginal Effects

Recall from Lecture 9: the Probit average marginal effect is a **nonlinear function** of $\hat{\beta}$:

$$\widehat{\text{AME}}_j = \frac{1}{n} \sum_{i=1}^n \phi(X_i' \hat{\beta}) \hat{\beta}_j$$

Apply the delta method. Define $h(\beta) = \text{AME}_j(\beta)$. Then:

$$\text{Asy. Var}(\widehat{\text{AME}}_j) = \nabla h(\hat{\beta})' \cdot \widehat{\text{Var}}(\hat{\beta}) \cdot \nabla h(\hat{\beta})$$

where ∇h accounts for both the direct effect ($\hat{\beta}_j$) and the indirect effect through $\phi(X_i' \hat{\beta})$.

The same delta-method machinery handles AMEs, long-run elasticities, and any other smooth transformation of $\hat{\theta}$.

Consistency of OLS: The Question

What we just assumed. The delta method required $\hat{\beta}$ to be close to β for large n . Is it?

What we are about to show. Under the assumptions on the next slide,

$$\hat{\beta} \xrightarrow{P} \beta,$$

i.e. $\Pr(\|\hat{\beta} - \beta\| > \epsilon) \rightarrow 0$ for every $\epsilon > 0$.

Tools, all from Lecture 10:

- **WLLN**: sample moments converge in probability to population moments.
- **CMT**: continuous functions of convergent sequences converge.

The next few slides apply these to $\hat{\beta}$, written as a function of sample moments.

Target of Estimation

- We are after the projection coefficient $\beta = (\mathbb{E}[\mathbf{x}\mathbf{x}'])^{-1}\mathbb{E}[\mathbf{x}Y]$ from the projection model $Y = \mathbf{x}'\beta + e$
- We assume $(Y_i, \mathbf{x}_i)$ are i.i.d.
- And define $\mathbf{Q}_{XX} = \mathbb{E}[\mathbf{x}\mathbf{x}']$, assume is positive definite.
- We also need moments to exist, including 4th moments for asymptotic normality.

Proof Outline: Two Routes to Consistency

- **Route 1** (Method of Moments):
 - WLLN: sample moments converge to population moments
 - CMT: continuous functions of convergent sequences converge
 - OLS is a continuous function of sample moments
- **Route 2** (Algebraic):
 - Write $\hat{\beta} - \beta$ as a product of sample moments
 - Apply WLLN to each piece, CMT/Slutsky to combine

Both routes use the same tools from Lecture 10. Route 1 generalizes to nonlinear estimators (MLE, GMM); Route 2 gives an explicit formula for the sampling error.

OLS is a function of sample moments

- $\hat{\beta} = \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i'\right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i Y_i\right) = \hat{Q}_{XX}^{-1} \hat{Q}_{XY}$
- $\hat{Q}_{XX}$, $\hat{Q}_{XY}$ are sample moments

Apply WLLN

- If $(\mathbf{x}_i, Y_i)$ are iid, then $\mathbf{x}_i \mathbf{x}_i'$, $\mathbf{x}_i Y_i'$ are iid.
- Recall from Lecture 10: WLLN says that if θ_i are iid with finite variance,

$$\frac{1}{n} \sum_{i=1}^n \theta_i \xrightarrow{P} \mathbb{E}[\theta]$$
- $\hat{\mathbf{Q}}_{XX} \xrightarrow{P} \mathbf{Q}_{XX}, \quad \hat{\mathbf{Q}}_{XY} \xrightarrow{P} \mathbf{Q}_{XY}$

Continuous Mapping Theorem (CMT)

- Recall from Lecture 10: if $Z_n \rightarrow_p c$ and $g(u)$ is continuous at c , then $g(Z_n) \rightarrow_p g(c)$.
- CMT tells us that plims survive continuous functions.
- $\hat{\beta} = g(\hat{Q}_{XX}, \hat{Q}_{XY})$ is a continuous function (matrix inversion is continuous at nonsingular matrices).
- $\hat{\beta} = \hat{Q}_{XX}^{-1} \hat{Q}_{XY} \rightarrow_p Q_{XX}^{-1} Q_{XY} = \beta$

Proof of Consistency of OLS: Route 2, Step 1

$$\begin{aligned}
 \hat{\beta} &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} && \text{(Definition)} \\
 &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'(\mathbf{X}\beta + \mathbf{e}) && \text{(Assumption 1)} \\
 &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e} && \text{(Inverse prop.)} \\
 &= (\beta + (1/n)n(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e}) \\
 &= (\beta + (\frac{1}{n}(\frac{1}{n})^{-1}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{e})) && \text{(trick)} \\
 \hat{\beta} &= \beta + (\frac{1}{n}\mathbf{X}'\mathbf{X})^{-1})(\frac{1}{n}\mathbf{X}'\mathbf{e})
 \end{aligned}$$

Proof of Consistency of OLS: Route 2, Step 2

$$\begin{aligned}
 \hat{\beta} &= \beta + \left(\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1}\left(\frac{1}{n}\mathbf{X}'\mathbf{e}\right) \\
 \text{plim}\hat{\beta} - \beta &= \left(\text{plim}\frac{1}{n}\mathbf{X}'\mathbf{X}\right)^{-1}\text{plim}\frac{1}{n}\mathbf{X}'\mathbf{e} && \text{(Slutsky)} \\
 &= \mathbf{Q}_{\mathbf{X}\mathbf{X}}^{-1}\mathbb{E}[\mathbf{X}\mathbf{e}] \\
 &= \mathbf{Q}_{\mathbf{X}\mathbf{X}}^{-1}\mathbf{0}
 \end{aligned}$$

The least squares estimator $\hat{\beta}$ is *consistent* for the projection coefficient β .

Key condition: Consistency requires $\mathbb{E}[\mathbf{x}\mathbf{e}] = \mathbf{0}$. When this fails (endogeneity), OLS is inconsistent — motivating instrumental variables (Lecture 13).

Asymptotic Normality: Algebraic Setup

Start from the sampling error decomposition:

$$\hat{\beta} - \beta = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i e_i \right) \quad (1)$$

Multiply both sides by $\sqrt{n}$:

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' \right)^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i e_i \right) \quad (2)$$

$$= \hat{Q}_{XX}^{-1} \left(\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i e_i \right) \quad (3)$$

Now we need the distribution of $\frac{1}{\sqrt{n}} \sum X_i e_i$. This is a sum of mean-zero iid random vectors — exactly the setting of the **multivariate CLT** from Lecture 10.

Asymptotic Normality: Applying CLT and Slutsky

$$\sqrt{n}(\hat{\beta} - \beta) = \underbrace{\hat{\mathbf{Q}}_{XX}^{-1}}_{\xrightarrow{p} \mathbf{Q}_{XX}^{-1}} \cdot \underbrace{\frac{1}{\sqrt{n}} \sum_{i=1}^n X_i e_i}_{\xrightarrow{d} N(0, \Omega)}$$

where $\Omega = \mathbb{E}[X_i X_i' e_i^2]$.

By Slutsky's theorem ($\xrightarrow{p} \times \xrightarrow{d}$):

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, \mathbf{Q}_{XX}^{-1} \Omega \mathbf{Q}_{XX}^{-1}) \equiv N(0, \mathbf{V}_{\beta})$$

Same decomposition as Probit MLE in Lecture 9: CLT for the numerator, WLLN/CMT for the denominator, Slutsky to combine.

Covariance of coefficient estimators under homoskedasticity

- Assume $Y = \beta_1 X_1 + \beta_2 X_2 + e$ with $E[ee'] = \sigma^2 I$.
- Suppose $E[X_1] = 0$, $E[X_2] = 0$, $Var[X_1] = 1$, $Var[X_2] = 1$, $corr[X_1, X_2] = \rho$

$$\mathbf{v}_{\beta}^0 = \sigma^2 \mathbf{Q}_{XX}^{-1} = \sigma^2 \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}^{-1} = \frac{\sigma^2}{1 - \rho^2} \begin{bmatrix} 1 & -\rho \\ -\rho & 1 \end{bmatrix}$$

- If X_1 and X_2 are positively correlated, $\hat{\beta}_1$ and $\hat{\beta}_2$ are negatively correlated.

Heteroskedastic Covariance Matrix Estimation

- The asymptotic covariance matrix of $\sqrt{n}(\hat{\beta} - \beta)$ is $\mathbf{Q}_{XX}^{-1}\Omega\mathbf{Q}_{XX}^{-1}$.
- This is the **sandwich form** from Lecture 6 — now with its asymptotic justification.

Estimator:

$$\hat{\Omega} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' \hat{e}_i^2$$

$$\hat{\mathbf{Q}}_{XX} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i'$$

$$\hat{\mathbf{V}}_{\beta}^{HC0} = \hat{\mathbf{Q}}_{XX}^{-1} \hat{\Omega} \hat{\mathbf{Q}}_{XX}^{-1}$$

This is the HC0 estimator. Recall from Lecture 6: HC1, HC2, HC3 apply finite-sample corrections. All are asymptotically equivalent.

Consistency of $\hat{\Omega}$

Add and subtract population errors:

$$\hat{\Omega} = \underbrace{\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' e_i^2}_{\xrightarrow{P} \Omega \text{ (WLLN)}} + \underbrace{\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i' (\hat{e}_i^2 - e_i^2)}_{\xrightarrow{P} 0}$$

The remainder vanishes: $\hat{e}_i - e_i = \mathbf{X}_i'(\beta - \hat{\beta}) \xrightarrow{P} 0$ by consistency of $\hat{\beta}$.

Therefore $\hat{\Omega} \xrightarrow{P} \Omega$, and by CMT:

$$\hat{\mathbf{V}}_{\beta}^{HC0} = \hat{\mathbf{Q}}_{XX}^{-1} \hat{\Omega} \hat{\mathbf{Q}}_{XX}^{-1} \xrightarrow{P} \mathbf{Q}_{XX}^{-1} \Omega \mathbf{Q}_{XX}^{-1} = \mathbf{V}_{\beta}$$

The Sandwich Pattern

The same “bread–meat–bread” structure appears throughout this course:

Model	Bread	Meat (Ω)	Simplification
OLS	Q_{XX}^{-1}	$\mathbb{E}[XX'e^2]$	$\sigma^2 Q_{XX}^{-1}$ if homosk.
Probit MLE	$\mathcal{I}^{-1}$	$\mathbb{E}[s_i s_i']$	$\mathcal{I}^{-1}$ if correct spec.
GMM	$(G'WG)^{-1}G'W$	$\mathbb{E}[g_i g_i']$	—

- Under “ideal” conditions (homoskedasticity / correct specification), the sandwich simplifies.
- Under misspecification, we need the full sandwich — **robust SEs**.
- Same `sandwich::vcovHC()` in R handles all cases.

Variances: Exact vs Asymptotic

Object	Formula	When to use
Exact variance (homosk.)	$(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{D}\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1}$ $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$	Fixed X , any n $+E[ee'] = \sigma^2 I$
Asy. variance (homosk.)	$\mathbf{Q}_{XX}^{-1}\Omega\mathbf{Q}_{XX}^{-1}$ $\sigma^2\mathbf{Q}_{XX}^{-1}$	Random X , n large $+E[e^2 X] = \sigma^2$

- The exact variance conditions on X ; the asymptotic variance averages over X .
- For large n : $\frac{1}{n}\mathbf{X}'\mathbf{X} \approx \mathbf{Q}_{XX}$, so they agree.
- In practice, we always estimate using the sandwich (or its homoskedastic special case).

From F Tests to Wald Tests

- We saw we can calculate an F test by forming a matrix of restrictions.
- The F distribution result depended on the homoskedasticity and normality of the errors.
- In large samples, we can instead rely on the asymptotic behavior of random variables and calculate a **Wald statistic**.
- Many test statistics are re-flavored Wald distances. Given a null that $E[\mathbf{q}] = \boldsymbol{\theta}$:

$$W = (\mathbf{q} - \boldsymbol{\theta})' [\widehat{\text{Var}}(\mathbf{q} - \boldsymbol{\theta})]^{-1} (\mathbf{q} - \boldsymbol{\theta})$$

- $W \sim \chi_J^2$ if $\mathbf{q}$ is Normal and $\text{Var}(\mathbf{q} - \boldsymbol{\theta}) = \Sigma$
- $W \overset{a}{\sim} \chi_J^2$ with CLT on $\mathbf{q}$ and consistency for $\hat{\Sigma}$

Mahalanobis Distance: Intuition

- Suppose $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. We want to measure how unusual an observation $\mathbf{x}$ is.
- In the scalar case:

$$D = \frac{x - \mu}{\sigma} \sim \mathcal{N}(0, 1) \quad \Rightarrow \quad D^2 \sim \chi_1^2$$

- In the vector case: transform the data to make it standard normal:

$$\mathbf{z} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{x} - \boldsymbol{\mu}) \sim \mathcal{N}(0, \mathbf{I})$$

- Then the squared norm of $\mathbf{z}$ gives:

$$\|\mathbf{z}\|^2 = (\mathbf{x} - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) = D^2$$

- This is the Mahalanobis distance: it tells us how far $\mathbf{x}$ is from $\boldsymbol{\mu}$ in standardized units, accounting for correlations and scale.

Wald Statistic as a Distance

- The Wald test generalizes the standardized t -statistic to higher dimensions:

$$\text{Univariate: } \frac{\hat{\beta} - \beta_0}{\text{SE}(\hat{\beta})} \sim \mathcal{N}(0, 1) \Rightarrow \left(\frac{\hat{\beta} - \beta_0}{\text{SE}(\hat{\beta})} \right)^2 \sim \chi_1^2$$

- In multivariate form:

$$W = (\mathbf{q} - \boldsymbol{\theta})' \hat{\boldsymbol{\Sigma}}^{-1} (\mathbf{q} - \boldsymbol{\theta})$$

- If $\mathbf{q}$ is consistent and asymptotically normal:

$$\sqrt{n}(\mathbf{q} - \boldsymbol{\theta}) \xrightarrow{d} \mathcal{N}(0, \boldsymbol{\Sigma}) \Rightarrow W \overset{a}{\sim} \chi_J^2$$

- Interpretation: Wald tests ask whether the observed $\mathbf{q}$ is “far” from $\boldsymbol{\theta}$, accounting for sampling variability.

Matrix form of Restrictions

- Suppose we have k parameters β and q restrictions with $q \leq (k)$. (Note $q = J$ in Greene, here k includes the intercept)
- Let $\mathbf{R}$ be a $q \times (k)$ matrix.
- We can express our null hypothesis as $H_0 : \mathbf{R}\beta = \mathbf{r}$.
- For example in the prior example where $\beta_2 = 0$ and $\beta_3 = \beta_4$

$$\mathbf{R} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix}, \quad \mathbf{r} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \\ \beta_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Wald Test for Linear Restrictions

$$W = (R\hat{\beta} - r)'[R\hat{V}_{\hat{\beta}}R']^{-1}(R\hat{\beta} - r)$$

$$W \overset{a}{\sim} \chi_q^2$$

- If we use $\hat{V}_{\hat{\beta}} = s^2(\mathbf{X}'\mathbf{X})^{-1}$ (homoskedasticity + normality), then W/q is exactly the F statistic distributed $F_{q,n-k}$.
- Without normality, the CLT tells us W/q is approximately $F_{q,n-k}$.
- Without normality or homoskedasticity, use $\hat{V}_{\hat{\beta}}^{HC}$ (sandwich) and $W \overset{a}{\sim} \chi_q^2$.

The Wald test nests F . With robust SEs, only consistency + asymptotic normality required. *Single coef:* $t^2 = F$ (HW4 Q3d).

Wald Tests in R

```
library(sandwich); library(lmtest); library(car)

mod0 <- lm(prestige ~ income + education + women,
           data = Prestige)

# Wald test with robust SEs (chi-squared)
linearHypothesis(mod0,
  c("education = 0", "women = 0"),
  vcov = vcovHC, test = "Chisq")

# Equivalent: waldtest with nested models
mod1 <- lm(prestige ~ income, data = Prestige)
waldtest(mod1, mod0, vcov = vcovHC, test = "Chisq")
```

Both test $H_0 : \beta_{ed} = \beta_{women} = 0$ using the sandwich covariance.

Applying the Delta Method to OLS

Earlier we introduced the delta method in general: if $\hat{\theta}$ is asymptotically normal, then any smooth function $h(\hat{\theta})$ is also asymptotically normal.

Now we have the specific ingredients for OLS:

- $\hat{\beta}$ is asymptotically normal (proved in the previous sections)
- $\hat{\mathbf{V}}_{\hat{\beta}}$ is a consistent estimator of its covariance (sandwich or homoskedastic)
- For any differentiable $\theta = r(\beta)$, the gradient $\mathbf{R} = \frac{\partial r}{\partial \beta} \big|_{\hat{\beta}}$ gives:

$$s(\hat{\theta}) = \sqrt{\mathbf{R}' \hat{\mathbf{V}}_{\hat{\beta}} \mathbf{R}}$$

This works for **any** differentiable function of $\hat{\beta}$ — linear or nonlinear.

Let's work through three examples of increasing complexity.

Running Example: Log Wage Regression

- $\log(\text{wage}) = \beta_1 \cdot \text{education} + \beta_2 \cdot \text{experience} + \beta_3 \cdot \text{experience}^2/100 + \beta_4 + e.$
- We can calculate:
 - Percentage return to education:

$$\theta_1 = 100\beta_1$$

- Percentage return to experience for those with 10 years of experience:

$$\theta_2 = 100\beta_2 + 20\beta_3$$

- Experience level which maximizes expected log wages:

$$\theta_3 = -50\beta_2/\beta_3$$

Each is a function $\theta = r(\beta)$. The first two are linear; the third is nonlinear. All three use the same delta method formula.

Example: Return to Education (Linear)

- Percentage return to education:

$$\theta_1 = 100\beta_1$$

- $R = \begin{pmatrix} 100 \\ 0 \\ 0 \\ 0 \end{pmatrix}$

- Asymptotic standard error:

$$s(\hat{\theta}_1) = \sqrt{R' \hat{V}_{\hat{\beta}} R} = 100 \cdot \sqrt{\hat{\text{Var}}(\hat{\beta}_1)}$$

This is just rescaling — the SE scales linearly.

Example: Return to Experience at 10 Years (Linear)

- Percentage return to experience at 10 years:

$$\theta_2 = 100\beta_2 + 20\beta_3$$

- Gradient of θ_2 with respect to β :

$$\mathbf{R} = \begin{pmatrix} 0 \\ 100 \\ 20 \\ 0 \end{pmatrix}$$

- Asymptotic standard error:

$$s(\hat{\theta}_2) = \sqrt{\mathbf{R}' \hat{\mathbf{V}}_{\hat{\beta}} \mathbf{R}}$$

This is a linear combination of two coefficients — the SE depends on their covariance.

Example: Maximum Return to Experience (Nonlinear)

- Experience level which maximizes expected log wages:

$$\theta_3 = -50 \frac{\beta_2}{\beta_3}$$

- Gradient of θ_3 with respect to β :

$$\mathbf{R} = \begin{pmatrix} 0 \\ -50/\beta_3 \\ 50\beta_2/\beta_3^2 \\ 0 \end{pmatrix}$$

- Asymptotic standard error:

$$s(\hat{\theta}_3) = \sqrt{\mathbf{R}' \hat{\mathbf{V}}_{\hat{\beta}} \mathbf{R}}$$

This is nonlinear — $\mathbf{R}$ itself depends on β , estimated at $\hat{\beta}$.

The z-statistic

In asymptotic inference the t -statistic is called the z -statistic:

$$T(\theta) = \frac{\hat{\theta} - \theta}{s(\hat{\theta})} = \frac{\sqrt{n}(\hat{\theta} - \theta)}{\sqrt{\hat{V}_{\theta}}} \xrightarrow{d} N(0, 1).$$

Under $H_0 : \theta = \theta_0$, reject at level α if $|T| > z_{\alpha/2}$ (e.g. $z_{0.025} = 1.96$).

Regression Intervals

For the CEF $m(x) = x'\beta$, the target parameter $\theta = r(\beta) = x'\beta$ has gradient $\mathbf{R} = x$, so the delta method gives

$$s(\hat{\theta}) = \sqrt{\mathbf{x}' \hat{\mathbf{V}}_{\hat{\beta}} \mathbf{x}}.$$

A 95% CI for $\mathbb{E}[Y \mid X = x]$ is

$$x'\hat{\beta} \pm 1.96 \cdot \sqrt{\mathbf{x}' \hat{\mathbf{V}}_{\hat{\beta}} \mathbf{x}}.$$

Example: Prestige $\sim$ Education

$$\beta_{ed} = 5.4, \beta_{cons} = -10.7, \hat{\mathbf{V}}_{\hat{\beta}} = \begin{bmatrix} 13.52 & -1.18 \\ -1.18 & .11 \end{bmatrix}$$

at 10 years of education:

$$\mathbf{x}'\hat{\beta} \pm 1.96 \cdot \sqrt{\mathbf{x}'\hat{\mathbf{V}}_{\hat{\beta}}\mathbf{x}}$$

$$(1 \ 10) \begin{pmatrix} -10.7 \\ 5.4 \end{pmatrix} \pm 1.96 \cdot \sqrt{(1 \ 10) \begin{bmatrix} 13.52 & -1.18 \\ -1.18 & .11 \end{bmatrix} \begin{pmatrix} 1 \\ 10 \end{pmatrix}}$$

$$43.3 \pm 1.96 \cdot \sqrt{0.87} = 43.3 \pm 1.83$$

compare to 22 years of education: $107 \pm 1.96 \cdot \sqrt{14.8} = 107 \pm 7.5$

R: Sandwich SEs and Delta Method

```
# Model-based vs sandwich SEs
coeftest(mod0)
coeftest(mod0, vcov = vcovHC(mod0, type = "HC1"))

# CI for CEF at a point
x0 <- c(1, 5000, 12, 30)
se <- sqrt(t(x0) %*% vcovHC(mod0) %*% x0)
cat(sum(x0 * coef(mod0)), "+/-", 1.96 * se)

# Delta method for a nonlinear function
car::deltaMethod(mod0, "income / education")
```

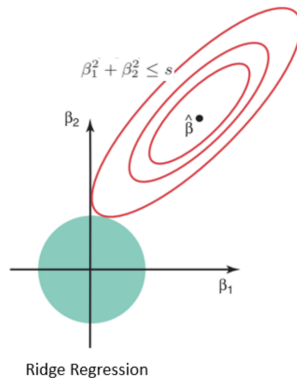
Application: Ridge Regression

- Suppose $Y = X'\beta + e$ with $\mathbb{E}[Xe] = 0$

$$\hat{\beta}_R = \left(\sum_{i=1}^n X_i X_i' + \lambda I_k \right)^{-1} \left(\sum_{i=1}^n X_i Y_i \right)$$

- Where $\lambda > 0$ is a constant, added to the diagonals of the covariance matrix.
- This is biased, shrinks $\hat{\beta}$.
- Arises from solving $\min_{\beta} (\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) + \lambda(\beta'\beta - c)$

Application: Ridge Regression



Ridge is Consistent (via CMT)

$$\begin{aligned}\text{plim } \hat{\beta}_R &= \text{plim} \left(\frac{1}{n} \sum_{i=1}^n X_i X_i' + \frac{\lambda}{n} I_k \right)^{-1} \left(\frac{1}{n} \sum_{i=1}^n X_i Y_i \right) \\ &= \left(\mathbb{E}[X_i X_i'] + \underbrace{\text{plim} \frac{\lambda}{n} I_k}_{\rightarrow 0} \right)^{-1} \mathbb{E}[X_i Y_i] \\ &= (\mathbb{E}[X_i X_i'])^{-1} \mathbb{E}[X_i Y_i] = \beta\end{aligned}$$

- As $n \rightarrow \infty$, Ridge converges to the same plim as OLS.
- Biased in finite samples, consistent asymptotically.

What We Proved Today

The OLS estimator $\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$ is:

Property	Tool (from Lectures 10–11)	Result
Consistent	WLLN + CMT	$\text{plim } \hat{\beta} = \beta$
Asy. normal	Multivariate CLT + Slutsky	$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(0, V_{\beta})$
Testable	Wald statistic	$W \overset{a}{\sim} \chi_q^2$
Robust	Sandwich estimator	$\hat{V}_{\beta}^{HC} \xrightarrow{p} V_{\beta}$

These results hold **without normality** — only requiring iid data with finite 4th moments and $\mathbb{E}[\mathbf{x}\mathbf{e}] = 0$.

Beyond OLS: the OLS Argument as a Template

Recap what we just did. In the normality step we wrote

$$\sqrt{n}(\hat{\beta} - \beta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \underbrace{\mathbf{Q}_{XX}^{-1} \mathbf{x}_i}_{\psi(Z_i)} e_i + o_p(1),$$

a sample mean of mean-zero random vectors plus a negligible remainder. CLT + Slutsky then delivered asymptotic normality.

The whole argument turned on two things:

- 1 Expressing $\sqrt{n}(\hat{\theta} - \theta)$ as a sample mean of an iid, mean-zero random quantity.
- 2 Controlling the leftover as $o_p(1)$.

Nothing about the argument was specific to OLS. Next slide: the general version.

Influence Functions (Preview)

The general template. Many estimators $\hat{\theta}$ admit the same representation:

$$\sqrt{n}(\hat{\theta} - \theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi(Z_i) + o_p(1), \quad \mathbb{E}[\psi(Z)] = 0.$$

The function ψ is the estimator's **influence function**: it measures how much a single observation Z_i moves $\hat{\theta}$ around θ .

CLT on the sample mean, Slutsky to absorb the $o_p(1)$:

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N(0, \text{Var}(\psi)).$$

HW4 Q4–Q9 builds this template for the AIPW estimator of the average treatment effect.

Looking Ahead

What happens when $\mathbb{E}[\mathbf{x}e] \neq 0$?

- OLS is **inconsistent**: $\text{plim } \hat{\beta} = \beta + Q_{XX}^{-1} \mathbb{E}[Xe] \neq \beta$
- No amount of data can fix this — the bias does not vanish.

Lectures 13–14 (IV and 2SLS): Find instruments Z with $\mathbb{E}[Ze] = 0$ and $\mathbb{E}[ZX'] \neq 0$.

- Replace the OLS moment condition $\mathbb{E}[X(Y - X'\beta)] = 0$ with $\mathbb{E}[Z(Y - X'\beta)] = 0$
- Same asymptotic machinery: WLLN, CLT, Slutsky, sandwich — applied to a different estimating equation
- This is another instance of the GMM framework: $\mathbb{E}[g(W_i, \theta_0)] = 0$